

TASK-2 REPORT

Cognifyz Machine Learning Internship

Project: Restaurant Recommendation System

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1. Objective

The goal of this task is to create a restaurant recommendation system.

The system suggests restaurants based on the user's preferred **cuisine, price range, and location** using a content-based filtering method.

2. Dataset

- **Dataset Name:** Restaurant.csv
- **Source:** Cognifyz Technologies

The dataset contains information about **9,551 restaurants**, including their names, locations, cuisines, price ranges, and ratings.

3. Method Used

- The dataset was loaded and cleaned
- Important columns such as cuisine, price range, location, and rating were selected
- Missing values were handled properly
- Text features were converted into numbers using **TF-IDF**
- **Cosine similarity** was used to find restaurants similar to user preferences

4. User Input and Output

The user enters:

- Cuisine
- Price range
- Location

The system then shows:

- Top 5 recommended restaurants
- Ratings of restaurants
- A short explanation for each recommendation

5. Results

- Restaurants matching the user's preferences are recommended
- Highly rated restaurants appear more often
- The explanation helps users understand why a restaurant was suggested

6. Conclusion

The restaurant recommendation system was successfully built using content-based filtering. It takes user input, processes restaurant data, and provides accurate and easy-to-understand recommendations.

7. Future Improvements

- Giving more importance to certain preferences
- Adding collaborative filtering
- Creating a web or app interface
- Sorting results by ratings